

Predicting Drug-Target Interactions Over Heterogeneous Information Network

Xiaorui Su , Pengwei Hu , *Member, IEEE*, Haicheng Yi , Zhuhong You , *Member, IEEE*, and Lun Hu , *Member, IEEE*

I. INTRODUCTION

Abstract—Identifying Drug-Target Interactions (DTIs) is a critical step in studying pathogenesis and drug development. Due to the fact that conventional experimental methods usually suffer from high costs and low efficiency, various computational methods have been proposed to detect potential DTIs by extracting features from the biological information of drugs and their target proteins. Though effective, most of them fall short of considering the topological structure of the DTI network, which provides a global view to discover novel DTIs. In this paper, a network-based computational method, namely LG-DTI, is proposed to accurately predict DTIs over a heterogeneous information network. For drugs and target proteins, LG-DTI first learns not only their local representations from drug molecular structures and protein sequences, but also their global representations by using a semi-supervised heterogeneous network embedding method. These two kinds of representations consist of the final representations of drugs and target proteins, which are then incorporated into a Random Forest classifier to complete the task of DTI prediction. The performance of LG-DTI has been evaluated on two independent datasets and also compared with several state-of-the-art methods. Experimental results show the superior performance of LG-DTI. Moreover, our case study indicates that LG-DTI can be a valuable tool for identifying novel DTIs.

Index Terms—Drug-Target interaction, DTI prediction, heterogeneous network embedding, semi-supervised learning.

DRUG-TARGET Interactions (DTI) play an important role in drug discovery and development [1]. Previous studies have reported that there are approximately 6000-8000 targets of pharmacological interest existed in the human genome [2], [3]. Since the identification of these targets facilitates the development of corresponding effective drugs, biologists and pharmacologists have made a lot of efforts in the past decades. However, only a few of them have been confirmed to be associated with approved drugs till now, thereby greatly limiting the understanding of biological processes and the development of novel drugs. To narrow such a huge gap, the development of computational methods proposed for DTI prediction has attracted much more attention in recent years, especially when several easy-to-access DTI datasets [4], [5], [6] are published. In addition, to overcome the disadvantages of time-consuming and high costs encountered in the traditional biological experiments, these computational methods also provide reliable evidence for scientists to accelerate drug discovery by identifying more accurate candidate targets.

Existing DTI prediction methods can be generally categorized into three groups by following [7]: (i) ligand structure-based, (ii) protein structure-based, and (iii) machine learning-based. Assuming that small molecular compounds with similar structures or physicochemical attributes tend to act on similar targets [8], methods in the first group mainly target to search chemical similarity and reverse pharmacophore when solving the DTI prediction problem. However, they only take into account the three-dimensional structures of biological targets while ignoring the associations between drugs and targets. In the second group, computational methods often make use of reverse molecular docking to select the best candidate combination as the potential target. As the first attempt, INVDOCK [9], developed by the National University of Singapore, has been successfully applied to drug molecular prediction. Another example in the second group is TarFisDock [10], which is an automatic search tool. TarFisDock has successfully been applied to identify the target peptide deformylase of natural products against *Helicobacter pylori*, and also determined the crystal structure of the complex composed of natural products and protein targets [11]. There is also another reverse-targeting tool called IDTarget [12] that predicts the interactions between small molecules and targets. Although protein structure-based methods can effectively identify drug targets, they could not be widely applicable, as the

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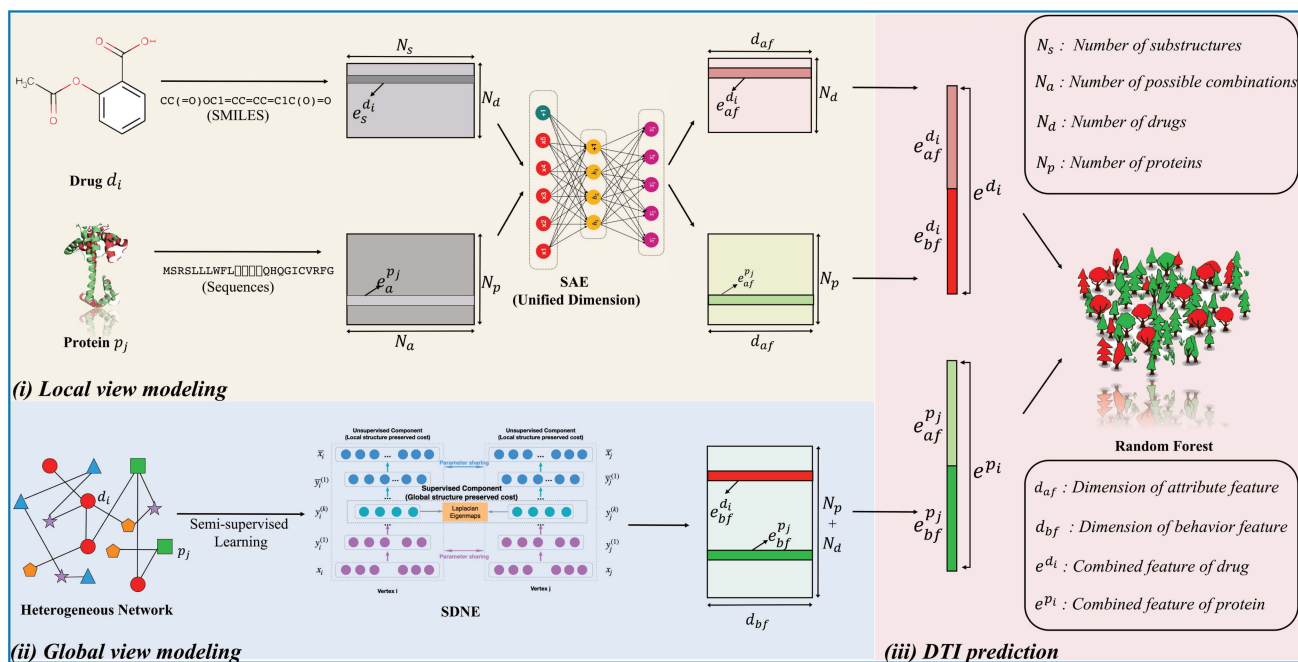


Fig. 1. An illustration of the complete procedure of LG-DTI.

three-dimensional structures of real targets are required to be known in advance, but only a few of them have been experimentally verified.

To address the above limitations, machine learning-based methods have been extensively studied in the field of DTI prediction. For example, Cheng et al. [13] regard the DTI prediction task as a recommendation problem, and apply supervised inference methods to predict drug targets from the DTI network. In spite of its good performance, the biological information of proteins is not incorporated into it. For further improvement in DTI prediction, Wang et al. [14] develop a computational model to encode drug structures and protein sequences as the feature vectors of drugs and targets respectively, and then predict potential DTIs with a rotation forest classifier. In addition, Huang et al. [15] also report a computational model for DTI prediction, and they characterize drug-protein pairs by combining the features of drugs and targets extracted from drug substructures and protein sequences. All aforementioned methods solve the task of DTI prediction by using the biological information of drugs and targets, however, they still fall short of handling the topological structure of DTI networks for improved accuracy. Since the structure of the DTI network reveals the mechanism of how drugs act on target proteins from a global perspective, it is essential to construct a more comprehensive model to predict DTIs by combining global network structure and local biological information of drugs and targets.

In this paper, we propose a novel network-based computational method, namely LG-DTI, to predict DTIs over heterogeneous information network (HIN) [16], which is designed to model a DTI network equipped with the biological information of drugs and targets. In particular, LG-DTI first learns the local

representations of drugs and targets from their molecular structures and protein sequences, and then adopts a semi-supervised network embedding model to obtain their global representations from the topological structure of HIN. With a concatenation aggregation function, these two kinds of features consist of the final representations of drugs and targets, which are then incorporated into a Random Forest classifier to complete the task of DTI prediction. To evaluate LG-DTI, experiments have been conducted on two practical datasets of different sizes, and the results demonstrate its superior performance in predicting DTIs. Our case study also indicates that LG-DTI can be a useful tool to discover novel DTIs.

The rest of this paper is organized as follows. The details of LG-DTI are described in Section II, following which the experimental results are presented and discussed in Section III. A conclusion is made in Section IV.

II. METHODOLOGY

Given a HIN composed of DTIs, drug molecular structures and target protein sequences, LG-DTI models the DTI prediction problem from the perspectives of local biological information and global network structure. The overall pipeline of LG-DTI is illustrated in Fig. 1. In particular, LG-DTI consists of three steps: (i) Local view modeling, which aims to learn attribute features from biological information; (ii) Global view modeling, which tends to learn behavior features of drugs and proteins from the topological structure of HIN; (iii) DTI prediction, which estimates the interaction probability for drug-target pairs by incorporating their attribute and behavior features with a Random Forest classifier.

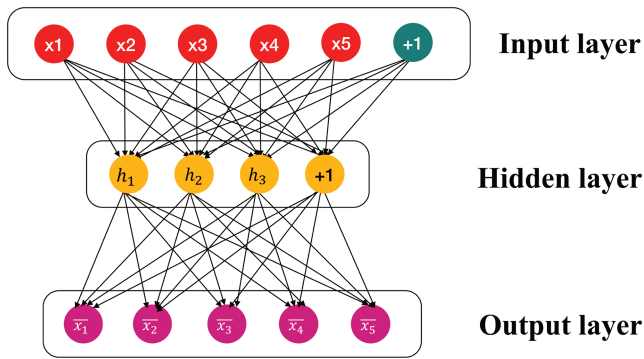


Fig. 2. An illustration of Stacked Autoencoder.

A. Local View Modeling

1) *Biological Feature Extraction of Drugs*: As one of the most commonly used expressions of drugs, Simplified Molecular Input Line Entry Specification (SMILES) [17] provides detailed information for the functions, structures, and targets of drugs. To extract the features of drugs from SMILES, we first download it from a widely used comprehensive drug database, i.e., DrugBank [18], then use Babel chemistry toolbox [19] to convert SMILES strings into molecular fingerprints, each bit of which indicates the existence of a particular substructure in the compound. For an arbitrary drug, each position of its fingerprint is set as 1 if there is a corresponding substructure in its SMILES, and 0 otherwise. Since there are 881 substructures in total, the drug fingerprint obtained can be viewed as an 881-dimensional vector.

2) *Biological Feature Extraction of Targets*: In our work, the biological features of drugs are extracted from their sequences, which have been proved to be an effective way to describe proteins in a variety of bioinformatics applications [20], [21], [22]. We first download the protein sequences from a well-established database, i.e., STRING [23], and then apply the 3-mer method to extract biological features from acquired sequences. According to the polarity of the side chain, there are 20 amino acids in total and they can be classified into four groups [20]. Hence, a sliding window with a fixed size of 3 is set to scan the whole protein sequence at a step size of 1, and the subsequence in the sliding window is recorded to update its number of occurrences in the corresponding position of a 64-dimensional feature vector. For an arbitrary target, each element in its feature vector indicates the frequency of the corresponding subsequence after normalization.

After obtaining the biological features of drugs and targets, stacked autoencoder [24] is applied to unify the dimensions of their feature vectors. The stacked autoencoder used here is composed of three layers as shown in Fig. 2, where encoder process can be formulated as: $h = \sigma(\mathbf{W}x + \mathbf{b})$. Then the decoder process is defined as: $\bar{x} = \sigma(\bar{\mathbf{W}}h + \bar{\mathbf{b}})$, where $\mathbf{W}$ and $\bar{\mathbf{W}}$ denote the weight matrices, $\mathbf{b}$ and $\bar{\mathbf{b}}$ represent the bias vectors, and σ denotes the activation function. Moreover, x is the biological feature vector of any drug or target, and $\bar{x}$ denotes its final unified feature vector with a fixed length.

TABLE I

CLASSIFICATION OF NETWORK REPRESENTATION LEARNING ALGORITHMS

Category	Method
Matrix Factorization (MF)-based	Laplacian [26], Grarep [27]
Random Walk (RW)-based	DeepWalk [28], Node2vec [29]
Neural Network (NN)-based	LINE [30], SDNE [31], GNN [32]

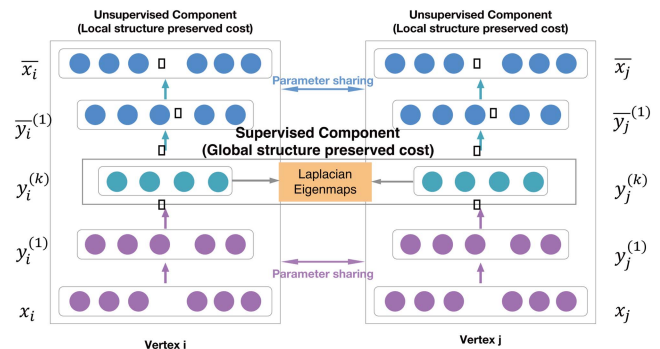


Fig. 3. An illustration of SDNE.

B. Global View Modeling

In addition to the biological information of drugs and targets, the topological structure of HIN is also important in modeling the DTI prediction task from a global perspective. In general, the topological information of nodes is obtained through the network representation algorithm. At present, three kinds of network embedding algorithms are widely used in the field of bioinformatics [25], as shown in Table I.

MF-based methods project the high dimensional adjacency matrix into lower dimensional matrices, while the such operation is not suitable for large-scale networks, which leads to large memory consumption and low-efficiency [33]. RW-based methods model the task by converting graph structure data into sequences through random walk, while the interactions between nodes in the graph are often more complex than node sequences, which makes that the structural information cannot be fully learned and preserved. Unlike the above two kinds of methods, SDNE models the network in a semi-supervised way by using auto-encoder combined with Laplacian Eigenmaps [26] simultaneously to build a multi-layer nonlinear function depth model for the nonlinear network structure [34]. Hence, as the first deep learning-based network representation learning method, SDNE [31], is adopted to learn the representations of drugs and proteins from the network structure perspective in our work.

Specifically, SDNE learns the behavior features of drugs and targets by jointly optimizing both first-order and second-order proximities in a semi-supervised deep learning model as shown in Fig. 3. The first-order proximity is trained by supervised learning to preserve the direct connections in HIN, while the second-order proximity is trained by unsupervised learning to capture the higher-order structure. As a result, for drugs and targets, their acquired feature vectors can preserve both lower- and higher-order network structures, and also are robust to sparse networks [35].

TABLE II
THE STATISTICS OF TWO DATASETS USED IN THE EXPERIMENTS

		MAN	OGB-biokg	E	NR
DTI	#Drugs	1,025	10,533	445	54
	#Proteins	1,649	17,499	664	26
	#Interactions	11,107	130,998	2,926	90
Network	#Nodes	6,528	93,773	1,109	80
	#Links ¹	105,546	5,088,434	11,086	496
	#Density ²	0.4953%	0.1157%	1.8028%	15.5%

¹ It should be noted that the number of links covers that of DTI interactions.

² Density is defined as $\frac{2 * \text{Links}}{\text{Nodes}^2}$

C. Drug-Target Interaction Prediction

After extracting the feature vectors of drugs and targets from local biological information and global network structure, LG-DTI addresses the task of DTI prediction by jointly considering these two kinds of features with an aggregation function. For an arbitrary node v in HIN, assuming that its local and global feature vectors are denoted as e_{af}^v and e_{bf}^v respectively, its final feature vector is formulated as:

$$e^v = \sigma(W_c \cdot (e_{af}^v; e_{bf}^v) + b_c), \quad (1)$$

where σ denotes the activation function, W_c and b_c are trainable parameters.

Since the task of DTI prediction can be regarded as a binary classification task, LG-DTI adopts a machine learning-based classifier to predict the interaction for a given drug-protein pair according to their final feature vectors. Regarding the issue of classifier selection, an in-depth discussion about the performance of LG-DTI given different classifiers is presented in Section of Ablation Study.

III. EXPERIMENTS

A. Dataset

To evaluate the performance of LG-DTI¹, four practical HINs, i.e., MAN, OGB-biokg, E, and NR, with different sizes are used in the experiments, and their statistic information is shown in Table II.

MAN [36] is a heterogeneous molecular network constructed by five kinds of molecules (drug, miRNA, lncRNA, disease, and protein), and their interactions. Moreover, all interactions in MAN have been validated by several well-established databases, such as HMDD [37], miRTarBase [38], lncRNADisease [39], STRING [23], DrugBank [18], CTD [40], DisGeNET [41], and lncRNA2Target [42]. There are 11,107 DTIs between 1,025 drugs and 1,649 targets.

OGB-biokg [43], constructed by Stanford University, is a heterogeneous biomedical network, which also contains five types of molecules (diseases, proteins, drugs, side effects, and proteins), and their interactions. There are a total of 10,533 approved drugs, 17,499 proteins, and 130,998 verified DTIs in OGB-biokg.

E [44], is a benchmark dataset to evaluate the model performance in DTI predictions, composed of drug nodes, target

nodes, and three kinds of corresponding interactions, such as drug-drug interactions, target-target interactions, and DTIs. The type of target proteins is all enzymes (E). There are 445 approved drugs, 664 proteins, and 2,926 verified DTIs in E.

NR [44], is also a benchmark dataset in the field of DTI predictions. Compared with the E dataset, the target proteins in this dataset are all nuclear receptors (NR). There are 90 DTIs between 54 drugs and 26 proteins.

B. Baselines

To demonstrate the accuracy of LG-DTI, we compare it with five state-of-the-art computational methods proposed for DTI prediction, including LINE-DTI, DTIP, NGDTP, KenDTI and DTIPred. The details of these baseline methods are given below.

- **LINE-DTI** [45]: predicts potential DTIs by combining the attribute and topological information of DTI network. One should note that the topological information is processed by Large-scale Information Network Embedding (LINE) [30].
- **DTIP** [46]: exploits the learning ability of convolutional neural networks to obtain 1D representations from protein sequences and SMILES strings, then takes them as the input of a fully connected neural network for DTI prediction.
- **NGDTP** [47]: proposes a network representation learning method based on matrix factorization. It integrates different kinds of protein and drug similarities to learn low-dimensional representations of drugs and proteins, then predicts potential DTIs by using gradient boosting decision tree.
- **KenDTI** [48]: utilizes both biochemical characteristics of drug molecular structures and protein sequences via word embeddings, and these characteristics are taken as the input for a convolutional neural network to learn the high-level representation of drugs and proteins for DTI prediction.
- **DTIPred** [49]: first learns the topological representation from a heterogeneous network through a random walk with restart and a convolutional neural network. After that, the original representations of drug-protein pairs are obtained by integrating multiple similarity matrices and DTIs.
- **DTI-CDF** [50]: is a similarity-based model, predicting DTIs based on drug similarities, target similarities, and drug-target pair similarities through the cascade deep forest.
- **NerLTR-DTA** [51]: uses the neighbor relationship of similarity and sharing to extract features, and applies a ranking framework with regression attributes to predict affinity values and priority order of query drug (or query target) and its related proteins (or compounds). It should be noted that when evaluating the performance of NerLTR-DTA in the experiments, we apply it as a binary classifier.

C. Experiment Settings

We first exclude all positive DTIs from the above datasets because the heterogeneous network should not contain any explicit information about the interactions between drugs and targets.

¹[Online]. Available: <https://github.com/blair1213/LG-DTI>

TABLE III
PERFORMANCES OF LG-DTI AND BASELINE MODELS UNDER FIVE-FOLD CV

Data.	Methods	Acc.	Sen.	Spe.	Pre.	AUC	AUPR
MAN	LINE-DTI	0.8199±0.0028	0.7403±0.0044	0.8074±0.0044	0.7996±0.0048	0.9014±0.0028	0.9076±0.0030
	DTIP	0.8078±0.0051	0.7736±0.0062	0.8582±0.0062	0.8441±0.0071	0.8286±0.0030	0.8833±0.0032
	NGDTP	0.7772±0.0056	0.7367±0.0053	0.8177±0.0058	0.8017±0.0064	0.8542±0.0056	0.8490±0.0058
	KenDTI	0.7835±0.0038	0.7471±0.0057	0.8200±0.0056	0.8058±0.0043	0.8686±0.0045	0.8523±0.0043
	DTIPred	0.8148±0.0027	0.7642±0.0034	0.8654±0.0038	0.8503±0.0040	0.8629±0.0035	0.8804±0.0036
	DTI-CDF	0.8345±0.0264	0.7588±0.0170	0.8500±0.0362	0.8519±0.0334	0.9052±0.0204	0.9128±0.0204
	NerLTR-DTA	0.7902±0.0139	0.7566±0.0197	0.8238±0.0305	0.8120±0.0258	0.8652±0.0158	0.8810±0.0160
	LG-DTI	0.8368±0.0025	0.7900±0.0048	0.8835±0.0048	0.8715±0.0044	0.9142±0.0023	0.9209±0.0028
OGB-biokg	LINE-DTI	0.8570±0.0006	0.8819±0.0022	0.8400±0.0024	0.8321±0.0019	0.9345±0.0010	0.9271±0.0013
	DTIP	0.7449±0.0025	0.7516±0.0035	0.7418±0.0035	0.7384±0.0051	0.8005±0.0018	0.7774±0.0017
	NGDTP	0.8790±0.0012	0.8893±0.0032	0.8714±0.0035	0.8687±0.0028	0.8802±0.0016	0.9083±0.0018
	KenDTI	0.8253±0.0044	0.8477±0.0048	0.8113±0.0055	0.8028±0.0057	0.9121±0.0032	0.9163±0.0030
	DTIPred	0.8517±0.0027	0.8496±0.0028	0.8532±0.0022	0.8538±0.0029	0.9330±0.0025	0.9324±0.0026
	DTI-CDF	0.8678±0.0068	0.8350±0.0096	0.8006±0.0113	0.8867±0.0094	0.9371±0.0066	0.9368±0.0068
	NerLTR-DTA	0.8346±0.0037	0.7896±0.0038	0.8797±0.0054	0.8678±0.0054	0.9022±0.0050	0.9020±0.0053
	LG-DTI	0.8988±0.0008	0.8934±0.0030	0.9032±0.0031	0.8962±0.0043	0.9548±0.0009	0.9498±0.0014
E	LINE-DTI	0.8191±0.0030	0.7671±0.0120	0.7891±0.0087	0.6418±0.0043	0.8191±0.0031	0.8623±0.0020
	DTIP	0.7250±0.0030	0.8290±0.0116	0.6211±0.0133	0.6863±0.0090	0.7250±0.0103	0.8004±0.0068
	NGDTP	0.7615±0.0074	0.8916±0.0094	0.6313±0.0091	0.7075±0.0061	0.7615±0.0074	0.8266±0.0048
	KenDTI	0.8651±0.0066	0.9189±0.0140	0.8112±0.0172	0.8298±0.0115	0.8651±0.0058	0.8947±0.0031
	DTIPred	0.8649±0.0103	0.9367±0.0075	0.7931±0.0208	0.8194±0.0148	0.8649±0.0103	0.8939±0.0074
	DTI-CDF	0.9381±0.0076	0.9531±0.0111	0.9231±0.0125	0.8767±0.0111	0.9381±0.0076	0.9510±0.0059
	NerLTR-DTA	0.8879±0.0173	0.8227±0.0383	0.9231±0.0096	0.9262±0.0098	0.8879±0.0173	0.9288±0.0093
	LG-DTI	0.9521±0.0040	0.9682±0.0060	0.9360±0.0115	0.9382±0.0102	0.9521±0.0040	0.9611±0.0041
NR	LINE-DTI	0.8023±0.0211	0.8996±0.0141	0.7049±0.0414	0.7539±0.0262	0.8023±0.0211	0.8519±0.0135
	DTIP	0.8551±0.0203	0.9152±0.0083	0.7951±0.0361	0.8178±0.0270	0.8552±0.0203	0.8877±0.0146
	NGDTP	0.7660±0.0201	0.8412±0.0193	0.6907±0.0295	0.7315±0.0213	0.7660±0.0170	0.8261±0.0115
	KenDTI	0.8701±0.0213	0.9193±0.0122	0.8209±0.0413	0.8381±0.0318	0.8701±0.0213	0.8988±0.0163
	DTIPred	0.8111±0.0125	0.9179±0.0109	0.7042±0.0305	0.7568±0.0175	0.8111±0.0210	0.8579±0.0125
	DTI-CDF	0.8628±0.0260	0.8089±0.0516	0.8968±0.0108	0.8866±0.0119	0.9581±0.0260	0.9155±0.0168
	NerLTR-DTA	0.8966±0.0136	0.9281±0.0094	0.8650±0.0251	0.8735±0.0208	0.8966±0.0136	0.9188±0.0109
	LG-DTI	0.9260±0.0092	0.9517±0.0078	0.8947±0.0151	0.9011±0.0129	0.9259±0.0180	0.9398±0.0092

Best results are bolded.

For all datasets, we randomly divide all approved positive DTIs into training, validation, and testing sets in an 8:1:1 ratio, and randomly select negative samples by paring up drugs and targets whose interactions are not found in the positive samples. To avoid the imbalance issue, the size of negative samples is the same as that of positive samples in all phases. LG-DTI and all the above baseline methods are evaluated by applying them to our benchmark datasets under five-fold cross-validation (CV) tests. In the experiments, LG-DTI selects Random Forest (RF) [52] as its default classifier.

For performance evaluation, the average scores obtained by each prediction method are recorded. Moreover, several evaluation metrics are adopted to quantitatively measure the performance of LG-DTI, and they are Accuracy (Acc.), Sensitivity (Sen.), Specificity (Spe.), Precision (Pre.), Area Under Curve (AUC), and Area Under Precision-Recall (AUPR). Their definitions are given as:

$$Acc. = \frac{TN + TP}{FP + TP + FN + TN} \quad (2)$$

$$Sen. = \frac{TP}{FN + TP} \quad (3)$$

$$Spe. = \frac{TN}{FP + TN} \quad (4)$$

$$Pre. = \frac{TP}{TP + FP}, \quad (5)$$

where TN , TP , FN , and FP denote true negative, true positive, false negative and false positive, respectively.

During the experiments, we define the dimension of drug/protein local/global view features as 64, the batch size and learning rate in SDNE are 200 and 0.01, and the number of estimators used in RF is 500. Besides, all the parameters used by the above baseline methods are set with the values recommended by their original work and are deployed on the same working machine equipped with Intel Core I7 2.6 GHz and 16 GB RAM.

D. Results and Analysis

Experimental results are reported in Table III as shown, where we note that LG-DTI outperforms the baseline methods across all datasets and evaluation metrics with only a few exceptions. The reasons for the superior performance of LG-DTI are manifold. First, compared with DTIP and KenDTI which predict DTIs only by using drug compounds and protein sequences, LG-DTI is also capable of learning the representations of drugs and targets from the perspective of global network structure, which further enhances the representative ability of final feature vectors. Second, comparing LG-DTI with NGDTP and DTIPred that simultaneously consider both the DTI network and the biological similarities of proteins and drugs, we note that the biological features extracted by LG-DTI are more effective as indicated by the results discussed in the later section. Last, compared with LINE-DTI which uses LINE to learn the network

embeddings of drugs and targets, LG-DTI integrates a semi-supervised learning model, i.e., SDNE, to learn the higher-order structures of HIN, which has already been verified to be useful for network analysis [53], [54], [55].

According to the experimental results presented in Table III we note that DTI-CDF and NerLTR-DTA generally perform better than all the other prediction methods except LG-DTI on all four datasets. The cascade deep forest (CDF) adopted by DTI-CDF is constructed based on both Random Forest and XGBoost. In doing so, the diversity and complementary among the base learners of CDF improve the prediction performance. Moreover, owing to the fact that DTI-CDF adopts multi-layer strategies in predicting DTIs, the multilayer characteristics of CDF are more fully exploited when DTI-CDF extracts the features of diseases and targets, thus leading to its promising performance across all four datasets. Similar to DTI-CDF, NerLTR-DTA utilizes multiple additive regression trees, each of which learns the residual of the conclusion sum of its previous trees, to predict potential DTIs by approximating the objective function closer to the actual distribution of samples. It is for this reason that NerLTR-DTA yields better performance than those adopting linear or generative classifiers. When comparing LG-DTI with DTI-CDF and NerLTR-DTA, we note that LG-DTI outperforms NerLTR-DTA in all 6 cases and DTI-CDF in 4 out of all 6 cases across different evaluation metrics and datasets. The main reason for the superiority of LG-DTI is that both NerLTR-DTA and DTI-CDF capture the characteristics of DTIs by only considering drug/target similarities and neighbor similarities, while LG-DTI additionally takes into account the complex high-order structure information with the integration of SDNE. Although DTI-CDF performs better than LG-DTI in terms of Spe. and AUC on the NR dataset, this phenomenon only indicates that DTI-CDF shows a stronger ability in distinguishing negative samples in NR as indicated by higher Spe. and AUC scores. For the problem of DTI prediction, what we concern most is to perform an accurate DTI prediction. In this regard, LG-DTI is preferred over DTI-CDF as a useful tool for identifying novel DTIs.

Moreover, we also note that LG-DTI exhibits a strong ability to deal with large-scale HINs, as its performance on OGB-biokg is much better than that on either of MAN, E and NR. The reasons are two-fold. First, large HINs offers LG-DTI is able to extract sufficient biological features from large HINs, thus making it to better distinguish drugs and proteins from a local perspective. Second, rich higher-order structures in large HINs offer more valuable hints on characteristics of DTIs from a global perspective, and the integration of SDNE allows LG-DTI to effectively capture such information for improved performance of DTI prediction.

E. Results and Discussion on Realistic Scenarios

As mentioned above, we have used the traditional five-fold CV to evaluate LG-DTI and all baseline methods. However, although five-fold CV is capable of avoiding the model suffering from over-fitting to a certain extent, it may lead to some optimistic results, which are difficult to reproduce in real-world [56].

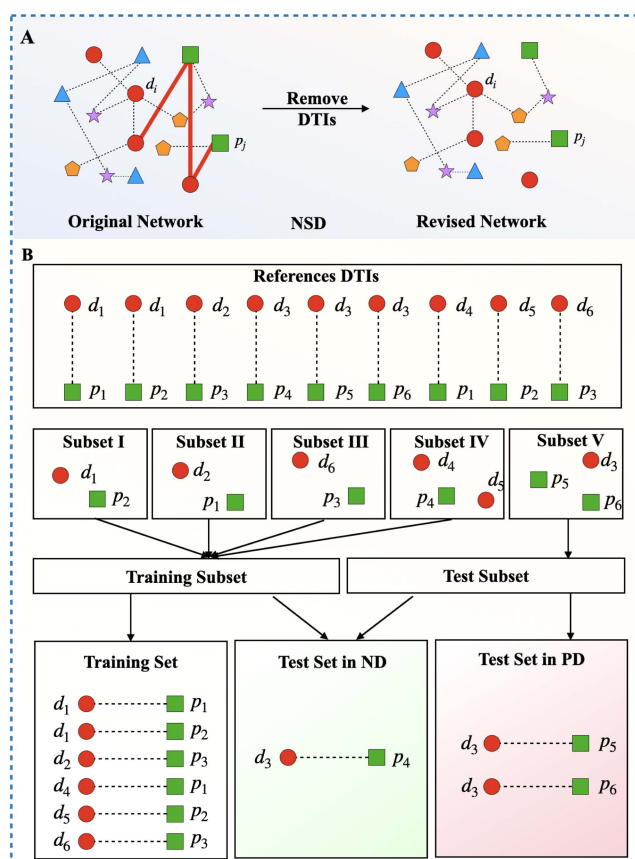


Fig. 4. The illustration of the procedure of three scenarios. (A) shows the network to be trained in the first NSD scenario; (B) shows the generation of test dataset in ND and PD scenarios, where only one fold and positive samples are illustrated.

To make a more realistic evaluation of the DTI prediction task, we create another three scenarios to further assess the performance of LG-DTI and baseline methods, and they are denoted as Neighbor-similarity Disjoint (NSD) [57], Node Disjoint (ND) [58], and Pairwise Disjoint (PD) [59], and their validation procedures are illustrated in Fig. 4. More specifically, the NSD scenario first composes a HIN by removing all DTIs from the training dataset and then adopts SDNE to learn the representations of drugs and diseases on that particular HIN. After that, all DTIs are randomly divided into 5 subsets of equal size. Each subset is alternatively used as the testing dataset while the rest are used to train the Random Forest classifier. By doing so, the bias resulting from neighbor similarities of drugs and diseases can thus be avoided for DTI prediction to a certain extent. In the ND scenario, dynamic drug-target pairs are generated. For each of them, there is only one node that is unseen in the training dataset while the other one is found in the training dataset. The purpose of using the ND scenario is to evaluate the performance of LG-DTI in terms of predicting dynamic drug-target pairs. The last scenario, i.e., PD, composes the testing dataset with drug-target pairs, each of which consists of nodes that are unseen in the training dataset. With this scenario, we aim to assess the ability of LG-DTI in processing unseen nodes. Figs. 5, Fig. 6,

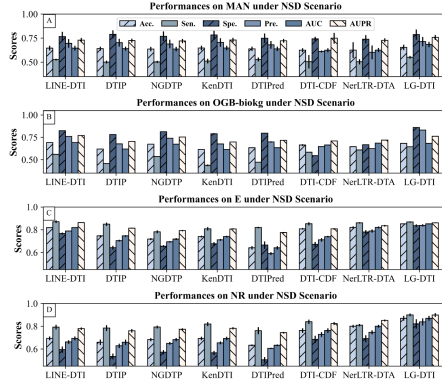


Fig. 5. Performances under NSD Scenario.

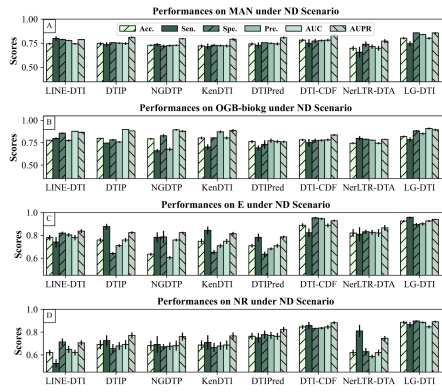


Fig. 6. Performances under ND Scenario.

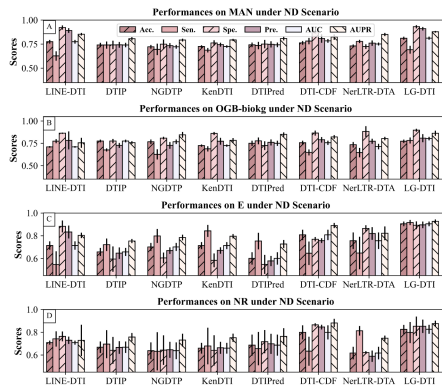


Fig. 7. Performances under PD Scenario.

and 7 report the results obtained by LG-DTI and all baseline algorithms in these three scenarios, where several things can be noted.

First, although the performances of LG-DTI and all baseline methods in all three scenarios degrade at different rates when compared with those under traditional five-fold CV, LG-DTI still achieves the best results across all scenarios. Second, due to the significant change in the network structure of HIN after removing all DTIs, the overall performance of LG-DTI falls by approximately 16%, 20%, 10%, and 15% on four datasets

respectively. Concerning the five-fold CV, all the drugs and targets in the testing fold involve almost 95% DTIs when we take their first-order neighbors into consideration. Consequently, the behavior features learned by LG-DTI under the five-fold CV can still reflect the global network structure of HIN even when 20% DTIs are removed. Hence, LG-DTI yields better performance under five-fold CV when compared with its performance in the NSD scenario. Last, the promising performance of LG-DTI in the ND and PD scenarios indicates the strong generalization ability of LG-DTI in processing unseen nodes in drug-target pairs.

F. Ablation Study

As mentioned above, nodes in a HIN can be represented in different ways, LG-DTI aggregates two representations in a concatenation way and adopts RF as the default classifier. In order to prove the effectiveness of integrated features, aggregation function, and the RF classifier, we conduct ablation studies on four datasets to investigate the impacts of biological features, aggregation functions, and classifier selections.

1) Impact of Biological Information and Network Structure:

In Table IV, it can be observed that the original version of LG-DTI yields the best performance when compared with its variants. Hence, the consideration of both biological information (LF) and network structure (GF) is an effective way for predicting DTIs over HIN. Compared with biological information, network structure is more representative, as it performs better across all evaluation metrics. This could be an indicator that the topological structure information of HIN is essential for DTI prediction. On the other hand, integrating both biological Information and network structure also leads to a more robust performance of LG-DTI as indicated by the results of standard deviation.

2) *Impact of Aggregation Function:* Besides concatenation function, there are still some widely used aggregation functions, such as SUM aggregator² and MEAN aggregator³. Table IV illustrates the results of different aggregators. It can be observed that the concatenation aggregator which has been integrated into LG-DTI achieves the best performance when compared with SUM and MEAN aggregators. The possible reason may be that the information will be lost to a certain extent when SUM or MEAN is adopted.

3) *Impact of Classifier Selection:* In order to investigate the impact of classifier selection, we replace RF with several different classifiers, including Logistic Regression (LR) [60], Naive Bayes (NB) [61], SVM [62], XGBoost [63] and AdaBoost [64], in LG-DTI, and evaluate their performances on four datasets. Experimental results are also presented in Table IV. Several things are worth noting.

First, we note that the performance of NB is the worst among all classifiers on all four datasets. A possible reason for that phenomenon is that as a generative model based on Bayesian

$$^2 e^v = \sigma(W_c \cdot (e_{af}^v + e_{bf}^v) + b_c)$$

$$^3 e^v = \sigma(\frac{1}{2} \cdot (e_{af}^v + e_{bf}^v))$$

TABLE IV
ABLATION STUDY RESULTS ON FOUR DATASETS

Data.	Var.	Acc.	Sen.	Spe.	Pre.	AUC	AUPR
MAN	LF	0.8117±0.0044	0.7698±0.0034	0.8537±0.0096	0.8403±0.0085	0.8797±0.0030	0.8863±0.0032
	GF	0.8272±0.0057	0.7778±0.0068	0.8766±0.0052	0.8631±0.0059	0.8963±0.0048	0.9081±0.0054
	SUM	0.7962±0.0155	0.7391±0.0213	0.8534±0.0310	0.8354±0.0285	0.8627±0.0217	0.8783±0.0165
	MEAN	0.7942±0.0093	0.7374±0.0145	0.8509±0.0140	0.8320±0.0130	0.8623±0.0082	0.8764±0.0105
	LR	0.7807±0.0071	0.7407±0.0084	0.8207±0.0068	0.8051±0.0074	0.8503±0.0032	0.8554±0.0045
	NB	0.6449±0.0096	0.3925±0.0427	0.8972±0.0449	0.7986±0.0485	0.7270±0.0208	0.7490±0.0223
	SVM	0.8108±0.0111	0.7462±0.0415	0.8755±0.0323	0.8587±0.0285	0.8565±0.0132	0.8764±0.0184
	XGBoost	0.8108±0.0062	0.7593±0.0059	0.8622±0.0069	0.8465±0.0074	0.8831±0.0056	0.8903±0.0058
	AdaBoost	0.7678±0.0068	0.7350±0.0096	0.8006±0.0113	0.7867±0.0094	0.8371±0.0066	0.8458±0.0069
	LG-DTI	0.8368±0.0025	0.7900±0.0048	0.8835±0.0048	0.8715±0.0044	0.9142±0.0023	0.9209±0.0028
OGB-biokg	LF	0.8609±0.0020	0.8861±0.0026	0.8436±0.0028	0.8683±0.0045	0.9397±0.0011	0.9336±0.0013
	GF	0.8720±0.0007	0.8654±0.0006	0.8770±0.0009	0.8892±0.0006	0.9390±0.0005	0.9397±0.0006
	SUM	0.8212±0.0013	0.8435±0.0020	0.8206±0.0026	0.8520±0.0026	0.8821±0.0013	0.8615±0.0013
	MEAN	0.7818±0.0013	0.8437±0.0021	0.8198±0.0031	0.7510±0.0029	0.8818±0.0013	0.8615±0.0013
	LR	0.5861±0.0015	0.5553±0.0148	0.5918±0.0025	0.6491±0.0051	0.6270±0.0013	0.6185±0.0012
	NB	0.5005±0.0011	0.9141±0.0036	0.5003±0.0006	0.0869±0.0025	0.5034±0.0007	0.7139±0.0044
	SVM	0.7759±0.0155	0.6764±0.0169	0.7754±0.0240	0.7270±0.0277	0.7759±0.0155	0.8576±0.0163
	XGBoost	0.8121±0.0017	0.7685±0.0034	0.8420±0.0018	0.8558±0.0020	0.8957±0.0008	0.9031±0.0009
	AdaBoost	0.7237±0.0032	0.6994±0.0057	0.7350±0.0031	0.7479±0.0039	0.8006±0.0022	0.8148±0.0025
	LG-DTI	0.8988±0.0008	0.8934±0.0030	0.9032±0.0031	0.8962±0.0043	0.9548±0.0009	0.9498±0.0014
E	LF	0.9041±0.0082	0.9398±0.0119	0.8683±0.0164	0.8774±0.0129	0.9041±0.0161	0.9236±0.0083
	GF	0.9321±0.0048	0.9593±0.0060	0.9049±0.0103	0.8656±0.0087	0.9321±0.0093	0.9448±0.0048
	SUM	0.9186±0.0042	0.9509±0.0027	0.8563±0.0094	0.8723±0.0071	0.9186±0.0075	0.9314±0.0042
	MEAN	0.9199±0.0092	0.9557±0.0087	0.8642±0.0234	0.8783±0.0176	0.9199±0.0092	0.9331±0.0075
	LR	0.7185±0.0083	0.7818±0.0128	0.6553±0.0096	0.6940±0.0071	0.7185±0.0083	0.7924±0.0059
	NB	0.6413±0.0159	0.9176±0.0067	0.5649±0.0329	0.5912±0.0122	0.6413±0.0159	0.7750±0.0061
	SVM	0.7401±0.0081	0.8084±0.0149	0.6717±0.0149	0.7113±0.0086	0.7401±0.0081	0.8078±0.0056
	XGBoost	0.9207±0.0093	0.9610±0.0057	0.8803±0.0172	0.8894±0.0141	0.9207±0.0093	0.9350±0.0074
	AdaBoost	0.7459±0.0067	0.8492±0.0145	0.6426±0.0168	0.7039±0.0081	0.7459±0.0067	0.8143±0.0045
	LG-DTI	0.9521±0.0040	0.9682±0.0060	0.9360±0.0115	0.9382±0.0102	0.9521±0.0040	0.9611±0.0041
NR	LF	0.8567±0.0609	0.8641±0.0942	0.8497±0.0897	0.8140±0.0611	0.8567±0.0609	0.8930±0.0439
	GF	0.8885±0.0435	0.8641±0.0442	0.8129±0.0892	0.8551±0.0565	0.8885±0.0434	0.8536±0.0295
	SUM	0.8888±0.0749	0.8676±0.0167	0.8100±0.0186	0.8801±0.0727	0.8888±0.0152	0.9093±0.0174
	MEAN	0.9002±0.0126	0.9098±0.0104	0.8908±0.0303	0.8468±0.0178	0.9002±0.0127	0.9108±0.0078
	LR	0.7825±0.0139	0.7931±0.0126	0.7719±0.0186	0.7495±0.0124	0.7825±0.0139	0.8730±0.0088
	NB	0.7330±0.0075	0.8612±0.0307	0.7047±0.0248	0.8059±0.0058	0.7330±0.0074	0.8433±0.0077
	SVM	0.8927±0.0155	0.8195±0.0174	0.7658±0.0179	0.8537±0.0123	0.8927±0.0155	0.9017±0.0102
	XGBoost	0.9057±0.0185	0.8342±0.0331	0.8773±0.0082	0.8566±0.0117	0.9058±0.0185	0.9318±0.0133
	AdaBoost	0.7594±0.0237	0.7843±0.0256	0.7346±0.0255	0.6275±0.0191	0.7584±0.0237	0.8099±0.0155
	LG-DTI	0.9260±0.0092	0.9517±0.0078	0.8947±0.0151	0.9011±0.0129	0.9259±0.0180	0.9398±0.0092

Best results of each dataset are bolded. LF represents local feature and GF denotes global feature.

theorem [65], NB assumes that individual attributes are independent of each other, and this assumption may be not consistent with the problem of DTI prediction. However, it obtains the highest Sen. on the OGB-biokg dataset, which may be mainly caused by data distribution.

Second, compared with NB, LR yields better performances on all four datasets. Taking Acc. as an example, LR is better by approximately 14% on MAN, 8.6% on OGB-biokg, 8.7% on E, and 4.9% on NR than NB. However, the performance of LR on OGB-biokg is not as well as it obtains on MAN. The major reason is that LR is a linear model, which is not capable of processing large HINs with high-dimensional features. Although SVM and LR both belong to the linear model, SVM performs better than LR on all four datasets, especially on OGB-biokg. Taking Acc. as an example, SVM improves the performance of LR by approximately 19%. The possible reason for this is that OGB-biokg is a large-scale but sparse dataset, which results in that LR is difficult to classify DTIs in the plane. However, SVM can map it to the separable space through kernel function for the solution.

Last, regarding the other three classifiers, i.e., XGBoost, AdaBoost, and RF, their performances are quite different despite the fact that all of them are ensemble-based models. Among them, RF yields the best performance as indicated by its largest scores and lowest standard deviations for all evaluation metrics. Hence, it is believed that RF should be considered the most suitable classifier for our task in terms of accuracy and robustness. When comparing XGBoost with AdaBoost, we note that XGBoost performs better on all datasets. The main reason is that XGBoost introduces regulations and a pruning strategy to better model the boundary of classification [66]. In sum, RF is selected as the recommended classifier of LG-DTI due to its promising performance.

G. Analysis of Heterogeneous Molecular Network

As mentioned above, the experiments have demonstrated that HINs contribute to the performance of DTI predictors. However, besides drug nodes, protein nodes and their interactions, other kinds of molecules, such as miRNA and lncRNA, and

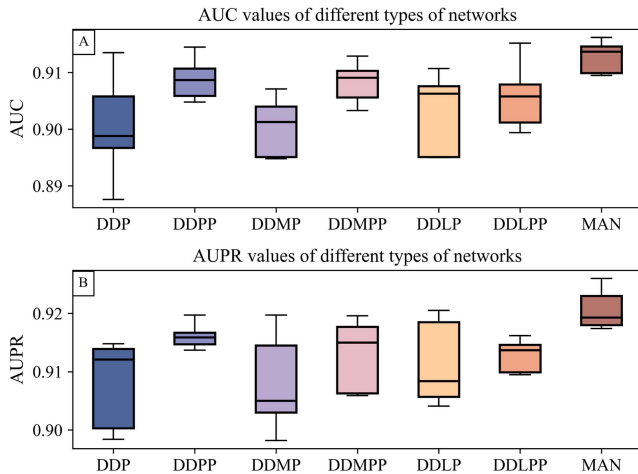


Fig. 8. Results obtained by six types of sub-networks. (A) shows the AUC values obtained by different sub-networks, while (B) compares the AUPR values of different sub-networks.

interactions between these nodes are also contained in the HINs used in this paper, which leads to that it difficult to determine which types of nodes/edges benefit to DTI predictions. To this end, we conduct related experiments on the MAN dataset by reconstructing sub-HINs from original networks according to biological knowledge.

According to the biological knowledge, we design six types of networks, which are denoted as DDP,⁴ DDPP,⁵ DDMP,⁶ DDMPP,⁷ DDLP,⁸ and DDLPP⁹ respectively. Fig. 8 illustrates the AUC and AUPR values obtained by the above six types of sub-networks. It can be found that: (i) the whole network contributes to DTI predictions most as its highest AUC and AUPR values; (ii) integration of protein-protein interactions (PPI) improves the performance of DDP, DDMP, and DDLP, respectively, which proves the importance of PPIs for DTI predictions [67].

H. Case Study

To further demonstrate the ability of LG-DTI in discovering novel DTIs, we pair up drugs and targets selected from DrugBank [18] and also ensure that these pairs are not found in both MAN and OGB-biokg. Top-20 drug-protein pairs with the largest prediction scores are listed in Table V and they are reckoned to be the most likely DTIs identified by LG-DTI. After a careful literature review, 16 of them have been confirmed by the latest version of KEGG [68]. Taking the first drug-target pair in Table V as an example, *Rosiglitazone* (D00596) is a thiazolidinedione, which is an adjunct to diet and exercise to maintain glycemic control in type 2 diabetes [69], and the target

TABLE V
THE PREDICTED DTIs WITH HIGH RANKS

Drug ID	Target ID	Rank	Evidence	Score
D00596	HSA: 5468	1	KEGG	0.9987
D03899	HSA: 7153	2	KEGG	0.9985
D07294	HSA: 5742	3	KEGG	0.9832
D08257	HSA: 3356	4	KEGG	0.9811
D00464	HSA: 2568	5	KEGG	0.9802
D08422	HSA: 1135	6	-	0.9785
D00550	HSA: 2556	7	KEGG	0.9742
D00550	HSA: 2559	8	KEGG	0.9731
D00066	HSA: 1543	9	-	0.9679
D01071	HSA: 2555	10	KEGG	0.9637
D00110	HSA: 6532	11	KEGG	0.9586
D00110	HSA: 6531	12	KEGG	0.9542
D00550	HSA: 2555	13	KEGG	0.9536
D07132	HSA:4986	14	KEGG	0.9501
D00064	HSA: 1576	15	-	0.9497
D09992	HSA: 1576	16	-	0.9485
D00790	HSA: 1813	17	KEGG	0.9450
D07294	HSA: 5743	18	KEGG	0.9434
D08489	HSA: 1815	19	KEGG	0.9420
D00513	HSA: 155	20	KEGG	0.9399

PPARG (HSA: 5468) is a nuclear receptor that binds to peroxisome proliferators, such as hypolipidemic drugs and fatty acids. Once activated by a ligand, *PPARG* binds to DNA-specific PPAR response elements (PPRE) and modulates the transcription of its target genes, such as acyl-CoA oxidase. Since the function of *PPARG* is to control the peroxisomal beta-oxidation pathway of fatty acids [70], [71], it can be a potential target of *Rosiglitazone*. Hence, LG-DTI is a useful tool for discovering novel DTIs.

IV. CONCLUSION

In this paper, we propose a novel network-based method, namely LG-DTI, to address the problem of DTI prediction over HIN. Different embedding strategies are adopted by LG-DTI to learn the respective representations of drugs and targets from local biological information and global network structure. After that, LG-DTI integrates a Random Forest classifier to predict DTIs by jointly considering these two kinds of features. Experimental results show the superior performance of LG-DTI when comparing it with five state-of-the-art methods in terms of several metrics. Our case study also indicates that LG-DTI is capable of discovering novel DTIs with considerable accuracy, as 16 out of the Top-20 drug-target pairs are confirmed by the latest version of KEGG. Regarding future work, we would like to model DTI prediction in an end-to-end fashion, expand the scale of the network by adding more related molecules, take link attributes like semantic relationships into consideration, and increase the interpretability of the model by introducing attention mechanism. We are also interested in investigating the influence of the negative sampling strategy for the purpose of generating negative samples with high quality.

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⁴Drug-Disease-Protein

⁵Drug-Disease-Protein-Protein

⁶Drug-Disease-miRNA-Protein

⁷Drug-Disease-miRNA-Protein-Protein

⁸Drug-Disease-lncRNA-Protein

⁹Drug-Disease-lncRNA-Protein-Protein

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